Network Security and Cryptography

2nd September 2024

Examination Paper

Answer ALL questions.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination

Answer ALL questions

Marks

Question 1

a) Explain the **THREE (3)** fundamental principles of network security. **6**

b) Alice wants to send a message to Bob using symmetric encryption. Describe what a **Message Authentication Code (MAC)** is and how Alice can use it to ensure the data integrity of the message received by Bob. **4**

Total 10 Marks

Question 2

a) Describe **TWO (2)** key differences between Block **and** Stream Ciphers. **4**

b) Describe **THREE (3)** methods utilised for sharing a secret key between two parties in symmetric cryptography. **6**

Total 10 Marks

Question 3

a) Explain the difference between Biometric Authentication **and** Multi-Factor Authentication (MFA). **4**

b) Compare and contrast discretionary access control (DAC), mandatory access control (MAC) and role-based access control (RBAC). **6**

Total 10 Marks

Question 4

a) State what password salting is **and** why is it crucial for password security. **4**

b) Explain what input validation is **and** why it is essential for preventing security vulnerabilities. **6**

Total 10 Marks

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Question 5

a) Discuss **THREE (3)** common characteristics of weak passwords and their security implications.. **6**

b) Explain **TWO (2)** key differences between a hash function **and** a one-time password (OTP). **4**

Total 10 Marks

Question 6

a) Explain what a buffer overflow is **and** how it can lead to significant software security vulnerabilities. **4**

b) Discuss **THREE (3)** recommendations you would propose to safeguard your computer system against malware. **6**

Total 10 Marks

Question 7

a) Describe **THREE (3)** strategies that can be implemented to prevent a ransomware attack from encrypting all data on a university network. **6**

b) Explain how DKIM (Domain Keys Identified Mail) contributes to email security. **4**

Total 10 Marks

Question 8

a) Explain what a firewall is **and** the difference between a hardware and software firewall. **6**

b) Discuss **ONE (1)** advantage and disadvantage of application-level proxy compared with packet filtering firewall. **4**

Total 10 Marks

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Question 9

a) Explain what a Virtual Private Network (VPN) is **and** the **TWO (2)** primary types of VPNs. **6**

b) Identify and explain **TWO (2)** potential risks associated with using public Wi-Fi networks to connect to a corporate network. **4**

Total 10 Marks

Question 10

a) Explain the difference between a host-based intrusion detection system (HIDS) and a network-based intrusion detection system (NIDS) **and** where each one would be more suitable on a network. **4**

b) Explain risk assessment **and** the difference between qualitative and quantitative risk assessments in information security. **6**

Total 10 Marks

End of paper

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